

IIT ROORKEE INTERNSHIP

Work Summary and Current Status

Spatial AI-Based Multimodal Water Quality Analysis and Prediction for the Ganga River
Kanpur–Varanasi Study Area

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Purpose of this document: This is a concise handover and project status document describing what has been implemented, the current state of the files and models, and what can be continued if the project is resumed later.

1. Project Overview

The project focused on developing a Spatial Artificial Intelligence framework for estimating water quality continuously along the Kanpur Varanasi stretch of the Ganga River.

The main challenge addressed was the spatial sparsity of conventional water quality monitoring stations. CWC observations were combined with environmental and anthropogenic spatial datasets to construct a continuous river-level dataset and subsequently train machine learning and Graph Neural Network models.

The completed workflow is:

CWC Data → Spatial Interpolation → Environmental Data Fusion → 500 River Nodes → Spatial Graph → ML / GraphSAGE → Unmonitored Location Inference

The project processed approximately 4,510 CWC records containing 53 parameters and generated a continuous 500-node representation of the study river stretch.

2. Work Completed

The following work has been completed during the internship.

1. Study Area and GIS Preparation

Defined the Kanpur Varanasi Ganga River study area using QGIS and shapefile boundaries. The working coordinate system was WGS84 (EPSG:4326).

2. CWC Water Quality Data Processing

Processed approximately 4,510 Central Water Commission water-quality records covering 53 parameters. Station-name inconsistencies were handled using coordinate-based filtering and spatial constraints.

3. Continuous River Geometry

Constructed a continuous river reference path and generated 500 spatial reference nodes along the river corridor.

4. Water Quality Interpolation

Interpolated water-quality parameters between available monitoring locations to obtain continuous target values at the 500 reference nodes.

5. Environmental Data Integration

Extracted spatial features from five additional datasets:

- CHIRPS – precipitation/rainfall
- GPWv4 – population density
- LULC – land-use/land-cover
- GRDC/CWC – river discharge
- GHSL – built-settlement proximity

6. Multimodal Dataset Construction

Merged the spatial features and interpolated water-quality targets into the master dataset:

`Ganga_Multimodal_Spatial_WQ.csv`

The final dataset contains 500 spatial nodes and 18 columns.

7. Spatial Graph Construction

Constructed a spatial k -nearest-neighbor graph over the 500 river nodes. The graph was subsequently converted into tensors for Graph Neural Network training.

8. Machine Learning Models

Implemented and evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Spatial GraphSAGE GNN

9. GraphSAGE Model

Implemented a two-layer spatial GraphSAGE architecture for aggregating information from neighboring river nodes.

10. Inference Pipeline

Developed an inference workflow capable of estimating multiple water-quality parameters at an unmonitored river location using its spatial and environmental features.

3. Current Results

The final reported benchmark results are shown below.

Table 1: Model Benchmark Results			
Model	MAE	RMSE	R^2
Linear Regression	0.6733	1.6767	0.8829
Random Forest	0.7328	1.9148	0.9782
XGBoost	0.7839	2.1082	0.9783
GraphSAGE	1.3273	3.5659	0.9373

Table 2: GraphSAGE Target-Wise Results

Parameter	MAE	RMSE	R^2
Dissolved Oxygen	0.2310	0.2816	0.8696
pH	0.0524	0.0638	0.9184
TDS	6.2686	7.9680	0.9712
Nitrate N	0.0567	0.0703	0.9655
Ammonia N	0.0276	0.0352	0.9621

For the GraphSAGE model, the target-wise results were:

The report records an example inference at an unmonitored coordinate (26.50°N, 81.20°E), where the pipeline predicted:

- DO = 6.9623 mg/L
- pH = 8.0727
- TDS = 228.4684 mg/L
- Nitrate N = 0.6283 mgN/L
- Ammonia N = 0.4008 mgN/L

4. Current File / Artifact Status

The following artifacts constitute the main project handover.

File / Directory	Purpose / Status
Ganga_Multimodal_Spatial_WQ.csv	Final 500 × 18 multimodal spatial dataset.
Ganga_Interpolated_Water_Quality.csv	500 spatial nodes with interpolated water-quality targets.
X.npy	Processed/scaled model feature matrix.
Y.npy	Processed target matrix.
edge_index.npy	Spatial k -NN graph connectivity.
water_quality_gcn.pth	Saved PyTorch GraphSAGE model weights.
build_spatial_dataset.py	Automated spatial/environmental dataset construction pipeline.
update_multimodal_dataset.py	Dataset alignment and update utility.
train_spatial_models.py	Model training, evaluation, and inference workflow.
01_..._ipynb - 09_..._ipynb	Modular notebooks covering data extraction, preprocessing, dataset construction, and modelling.
CWC/	Raw CWC water-quality data.
CHIRPS_DS/	Rainfall data.
GPW4/	Population-density raster data.
LULC_MONTHLY/	Land-use/land-cover data.
GRDC/	River discharge data.
GHSL_shapefile/	Settlement/build-up spatial data.

5. Present Status

- **Data collection and preprocessing:** Completed.
- **Study-area preparation:** Completed.
- **CWC station processing:** Completed.
- **Continuous 500-node river dataset:** Generated.
- **Environmental feature extraction:** Completed.
- **Multimodal dataset:** Generated and preserved.
- **Spatial graph:** Constructed.
- **Baseline ML models:** Implemented and evaluated.
- **GraphSAGE model:** Implemented, trained and evaluated.
- **Inference pipeline:** Implemented and tested on an example unmonitored location.
- **Documentation:** Technical internship report and supporting notebooks/scripts prepared.

Overall status: The core data-processing, multimodal fusion, modelling and inference pipeline has been implemented. The project is therefore at a stage where further work can begin from the existing datasets, scripts, notebooks and model artifacts rather than rebuilding the pipeline.

6. How to Continue the Work

If the project is resumed after a few months, the recommended starting point is:

1. Start with `Ganga_Multimodal_Spatial_WQ.csv` to inspect the final modelling dataset.
2. Review `build_spatial_dataset.py` for the environmental data extraction pipeline.
3. Review `train_spatial_models.py` for model training, benchmarking and inference.
4. Use `water_quality_gcn.pth` for the saved GraphSAGE model weights.
5. Check the notebooks 01–09 for the step-by-step experimental workflow.
6. Reproduce the existing benchmark before introducing a new model or dataset.

This provides a direct path from the preserved files to the existing results.

7. Possible Further Work

The following are natural extensions identified from the current project:

1. **Strict spatial validation:** Use spatial block holdout rather than only random train/test splitting to better evaluate generalization between geographically separated river segments.
2. **Temporal modelling:** Extend the static spatial model to a spatiotemporal GNN to capture temporal changes in water quality and environmental conditions.
3. **Hydrological graph topology:** Replace the simple spatial k -NN graph with actual river network topology, for example using HydroRIVERS flow connectivity.
4. **Remote sensing integration:** Add Sentinel-2 surface reflectance features such as B2, B3, B4 and B8 and derive water-related spectral indicators such as NDWI.
5. **Model interpretability:** Use SHAP to analyse the contribution of rainfall, population density, discharge, LULC and other predictors to water-quality predictions.
6. **Web GIS deployment:** Deploy the inference pipeline through an interactive GIS interface using Streamlit, Leaflet or a similar framework.

8. Important Technical Notes

A few limitations should be kept in mind when continuing the project.

- The continuous water-quality targets are based on spatial interpolation between sparse CWC monitoring stations and therefore represent spatial estimates rather than direct measurements at every generated node.
 - Random spatial train/test splitting can introduce spatial autocorrelation between nearby training and testing nodes. Therefore, the current benchmark should not be interpreted as a strict test of out-of-region generalization.
 - The environmental datasets have different spatial resolutions and temporal characteristics.
 - The current graph represents spatial proximity using k -NN rather than true river flow connectivity.
- These points should be considered before extending or comparing new models.

9. Technical Stack

Component	Technology
Programming	Python
GIS	QGIS
Spatial Processing	GeoPandas, Shapely, Rasterio
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost
Deep Learning	PyTorch
Graph Learning	GraphSAGE / PyTorch Geometric workflow
Visualization	Matplotlib / Seaborn
Data Formats	CSV, GeoTIFF, Shapefile, TXT, NPY, PTH

10. Handover Summary

Current handover point:

The project currently has a complete working data-to-model pipeline for the Kanpur–Varanasi Ganga River study. The processed multimodal dataset, interpolated water-quality dataset, spatial graph tensors, model weights, Python scripts and Jupyter notebooks have been preserved. A future researcher can therefore start from the existing master dataset or reproduce the pipeline from the raw datasets and preprocessing scripts. The main scope for continuation is improved spatial/temporal validation, better river-network graph modelling, additional remotesensing features, advanced GNN architectures and deployment.

Prepared as a project handover and current status summary

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